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Project Documentation

Human Resources Dataset Analysis

Workforce Insights through Power BI Dashboards

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Table of Contents

1. Project Overview	2
2. Data Description	3
2.1 Dataset at a Glance	3
2.2 Dataset Attributes	3
3. Problems of Data	4
4. Data Structure	5
4.1 Original Structure	5
4.2 Relational Model (After Normalisation)	5
5. Data Modelling	6
5.1 ETL Process	6
5.2 Dashboard Design	6
5.3 DAX Measures – Summary Table	7
5.3 DAX Measures – Key Code Definitions	8
6. Key Performance Indicators (KPIs)	9
7. Unavailable KPIs	10
8. Tools & Technologies	10
9. Key Findings & Recommendations	11

1. Project Overview

We built this project to understand what is happening inside a company's workforce — who is leaving, who is being paid fairly, and who might need more attention. Using an HR dataset with over 1,400 employee records, we loaded the data into Power BI, cleaned it up, and built dashboards that make it easy to spot patterns that would otherwise be buried in spreadsheets.

The work covered the full analytics pipeline: loading and cleaning data in Power Query, restructuring it into a proper relational model, writing DAX measures to calculate key figures, and designing dashboard pages that each answer a specific business question.

Platform: Microsoft Power BI Desktop | **File:** HR.pbix | **Source:** 5 CSV files (Employee, PerformanceRating + 3 dimension tables)

2. Data Description

2.1 Dataset at a Glance

The dataset covers employee records from a mid-sized organisation. Each row represents one employee, and the columns span everything from basic demographics to satisfaction scores and whether the employee eventually left the company. The raw data came across five separate CSV files, which we connected in Power BI using shared ID columns.

2.2 Dataset Attributes

Category	Columns Included
Demographics	Age, AgeGroup, Gender, MaritalStatus, State, Ethnicity, DistanceFromHome (KM), DistanceGroup
Employment	EmployeeID, Department, JobRole, BusinessTravel, HireDate, OverTime, Attrition
Education	Education (level 1–5), EducationField, EducationLevelID
Compensation	Salary, StockOptionLevel
Tenure	YearsAtCompany, YearsInMostRecentRole, YearsSinceLastPromotion, YearsWithCurrManager
Satisfaction	EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction, WorkLifeBalance
Performance	SelfRating, ManagerRating, TrainingOpportunitiesWithinYear, TrainingOpportunitiesTaken, PerformanceCategory

3. Problems of Data

Before we could start building dashboards, we went through the raw data and found several issues that would have caused incorrect numbers or broken visuals if left untreated:

- **Duplicate records:** Some employee IDs appeared more than once because of how the data was exported. We removed the extra rows using Power Query's Remove Duplicates function.
- **Inconsistent column naming:** Column headers mixed lowercase-with-underscores and PascalCase styles. We standardised all headers to PascalCase so joins between tables would work correctly.
- **Blank and null values:** A number of satisfaction columns had empty cells. Instead of deleting those rows, we replaced the blanks with the column median to avoid skewing averages.
- **Redundant constant columns:** Fields like EmployeeCount (always 1) and Over18 (always "Y") carried no useful information, so we dropped them to keep the model clean.
- **Flat file structure:** All the data came in one large table. We split it into separate dimension tables for Education, Satisfaction, and Rating levels to build a proper relational model.
- **Dates stored as text:** HireDate and ReviewDate were plain text strings. We converted them to the Date data type so time-based filters and calculations would work correctly.
- **Salary outliers:** A small number of salary values were unusually high. We applied the IQR method ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$) to define acceptable bounds and flagged the outliers rather than deleting them.

Note: All transformations were performed inside Power Query. Each step is recorded and visible in the Applied Steps panel of the HR.pbix file.

4. Data Structure

4.1 Original Structure

The raw data arrived as flat CSV files with no relationships between them. The main Employee file alone had over 1,400 rows and 23 columns, with department names and satisfaction values repeated on every single row — the typical problem with denormalised data.

4.2 Relational Model (After Normalisation)

We restructured the data into a relational model that matches what is visible in Power BI's Model View. The two core tables connect to four small dimension lookup tables, and a disconnected Measures table holds all DAX calculations.

Employee (Core Table)

EmployeeID (PK), Age, AgeGroup, Attrition, BusinessTravel, Department, DistanceFromHome (KM), DistanceGroup, Education, EducationField, Gender, HireDate, JobRole, MaritalStatus, OverTime, Salary, StockOptionLevel, YearsAtCompany, YearsSinceLastPromotion

PerformanceRating (Core Table)

PerformanceID (PK), EmployeeID (FK), ReviewDate, EnvironmentSatisfaction, EnvironmentSatisfactionStatus, JobSatisfaction, JobSatisfactionStatus, ManagerRating, PerformanceCategory, SelfRating, TrainingOpportunitiesWithinYear, WorkLifeBalance

Table	Type	Key Columns
EducationLevel	Dimension	EducationLevelID (PK), EducationLevel
SatisfiedLevel	Dimension	SatisfactionID (PK), SatisfactionLevel
RatingLevel	Dimension	RatingID (PK), RatingLevel
Date	Date Dimension (DAX)	Date, Month, Month Name, Quarter, Year
Measure	Measures Table	All 25 custom DAX measures (disconnected)

5. Data Modelling

5.1 ETL Process (Power Query)

We ran all data preparation steps inside Power Query before loading anything into the model:

- **Data Ingestion:** Imported all five CSV files via Get Data → Text/CSV. Each file was checked for correct column types on arrival.
- **Column Cleaning:** Dropped constant columns, renamed headers to PascalCase, and corrected any data type mismatches (dates, integers, text).
- **Duplicate and Null Handling:** Removed duplicate EmployeeIDs and replaced blank satisfaction scores with the column median using Replace Values.
- **Dimension Table Creation:** Used Reference queries to split the flat Employee table into separate dimension tables, adding surrogate primary keys where needed.
- **Relationship Setup:** In Model View, created one-to-many relationships from each dimension table's PK to the matching FK in the core tables. Cross-filter direction set to Single.
- **Date Table:** Created a calculated Date table using the DAX CALENDAR function and marked it as the official date table to enable time-intelligence functions.

5.2 Dashboard Design

The report is organised into five pages, each focused on a specific HR question:

- **Overview:** KPI cards — total employees, active count, attrition rate, average salary, average age, and average tenure.
- **Attrition Analysis:** Breakdown of attrition by department, job role, overtime status, and tenure band.
- **Compensation Analysis:** Average, median, max, and min salary; total payroll; outlier bounds via IQR.
- **Demographics and Diversity:** Gender ratio, age groups, education fields, and commute distance distribution.
- **Performance and Satisfaction:** Manager vs. self ratings, satisfaction levels by department, and training usage.

5.3 DAX Measures – Summary Table

A dedicated *Measure* table holds all 25 custom DAX measures used across the dashboards:

Measure Name	DAX Definition
WORKFORCE COUNTS	
Total Employee	DISTINCTCOUNT(Employee[EmployeeID])
Active Employee	CALCULATE(DISTINCTCOUNT(Employee[EmployeeID]),Employee[Attrition]="No")
Female employees	CALCULATE(DISTINCTCOUNT(Employee[EmployeeID]),Employee[Gender]="Female")
Male employees	CALCULATE(DISTINCTCOUNT(Employee[EmployeeID]),Employee[Gender]="Male")
Female %	DIVIDE([Female employees],[Total Employee])
Male %	DIVIDE([Male employees],[Total Employee])
Overtime workers	CALCULATE(DISTINCTCOUNT(Employee[EmployeeID]),Employee[OverTime]="Yes")
ATTRITION	
Attrition Count	CALCULATE(DISTINCTCOUNT(Employee[EmployeeID]),Employee[Attrition]="Yes")
Attrition Ratio	DIVIDE([Attrition Count],[Total Employee],0)
Avg Tenure Attrition	CALCULATE([Average YearAtComp],Employee[Attrition]="Yes")
AvgSalary Attrition	CALCULATE([Average Salary],Employee[Attrition]="Yes")
OverTime Attrition	CALCULATE([Attrition Count],Employee[OverTime]="Yes")
SALARY & COMPENSATION	
Average Salary	AVERAGE(Employee[Salary])
Median Salary	MEDIAN(Employee[Salary])
Max salary	MAX(Employee[Salary])
min salary	MIN(Employee[Salary])
Total Payroll	SUM(Employee[Salary])
Q1 Salary	PERCENTILE.INC(Employee[Salary],0.25)
Q3 Salary	PERCENTILE.INC(Employee[Salary],0.75)
IQR Salary	[Q3 Salary] - [Q1 Salary]
Lower Bound Salary	[Q1 Salary] - (1.5 * [IQR Salary])
Upper Bound Salary	[Q3 Salary] + (1.5 * [IQR Salary])
DEMOGRAPHICS & TENURE	
Average Age	AVERAGE(Employee[Age])
Average YearAtComp	AVERAGE(Employee[YearsAtCompany])
Avg years since promotion	AVERAGE(Employee[YearsSinceLastPromotion])

5.3 DAX Measures – Key Code Definitions

The definitions below show measures that combine multiple functions or reference other measures. Simple single-function measures are already shown in the table on the previous page.

ATTRITION COUNT & ATTRITION RATIO

```
Attrition Count =
    CALCULATE (
        DISTINCTCOUNT( Employee[EmployeeID] ),
        Employee[Attrition] = "Yes"
    )

Attrition Ratio = DIVIDE( [Attrition Count], [Total Employee], 0 )
```

ATTRITION DEEP-DIVE

```
Avg Tenure Attrition = CALCULATE( [Average YearAtComp], Employee[Attrition] = "Yes" )
AvgSalary Attrition = CALCULATE( [Average Salary], Employee[Attrition] = "Yes" )
OverTime Attrition = CALCULATE( [Attrition Count], Employee[OverTime] = "Yes" )
```

SALARY OUTLIER DETECTION – IQR METHOD

```
Q1 Salary = PERCENTILE.INC( Employee[Salary], 0.25 )
Q3 Salary = PERCENTILE.INC( Employee[Salary], 0.75 )

IQR Salary = [Q3 Salary] - [Q1 Salary]
Lower Bound Salary = [Q1 Salary] - ( 1.5 * [IQR Salary] )
Upper Bound Salary = [Q3 Salary] + ( 1.5 * [IQR Salary] )
```

GENDER SPLIT

```
Female employees = CALCULATE( DISTINCTCOUNT( Employee[EmployeeID] ),
    Employee[Gender] = "Female" )
Male employees = CALCULATE( DISTINCTCOUNT( Employee[EmployeeID] ),
    Employee[Gender] = "Male" )

Female % = DIVIDE( [Female employees], [Total Employee] )
Male % = DIVIDE( [Male employees], [Total Employee] )
```

6. Key Performance Indicators (KPIs)

These are the KPIs we were able to calculate from the available data. Each one maps to a real DAX measure in the Power BI report:

#	KPI Name	What It Measures	DAX Measure
1	Attrition Rate	Percentage of employees who left	Attrition Ratio
2	Avg. Monthly Salary	Mean salary across all employees	Average Salary
3	Avg. Tenure	Average years of service per employee	Average YearAtComp
4	Total Payroll	Sum of all employee salaries	Total Payroll
5	Gender Diversity Ratio	Male vs. female employee split	Male % / Female %
6	Overtime Rate	Share of workforce on overtime	Overtime workers ÷ Total Employee
7	Promotion Recency	Average years since last promotion	Avg years since promotion
8	Salary Range	Gap between highest and lowest salary	Max salary – min salary
9	Salary IQR	Middle 50% salary spread	IQR Salary
10	Attrition by Overtime	Number of leavers who were on overtime	OverTime Attrition

7. Unavailable KPIs

Some common HR metrics could not be calculated because the required data was not part of the dataset:

#	KPI Name	Reason It Is Missing
1	Revenue Per Employee	No financial revenue data in this HR dataset
2	Cost of Hiring	Recruitment cost records are not captured
3	Training Hours Per Employee	Only training count is available, not hours spent
4	Absenteeism Rate	No absence or leave records in the dataset
5	Time to Fill Positions	Recruitment timeline data is not included
6	Employee Net Promoter Score	eNPS survey data was not part of the source files

8. Tools & Technologies

Tool	How We Used It
Microsoft Power BI Desktop	Main platform — data model, DAX measures, dashboard pages
Power Query (M Language)	All ETL steps: cleaning, type corrections, table splits, key generation
DAX	Writing 25 custom measures stored in the Measure table
Microsoft Excel	Quick review and filtering of the raw CSV files before import

9. Key Findings & Recommendations

Based on the Power BI dashboard analysis, the following data-driven insights and strategic recommendations were developed:

Category	Key Finding (Insight)	Strategic Recommendation
Attrition Risk	16.12% attrition rate, heavily driven by overtime workers. Technology is the highest risk department.	Reduce overtime load in the Technology department to lower burnout and attrition risk.
Salary Disparities	Wide range (20K to 547K). Doctorate holders earn 63% more than those with no formal education.	Implement salary band reviews for entry-level roles—the gap between 20K and the median is large.
Gender Balance	Near-equal split (44.29% Male vs 45.92% Female). Non-binary employees make up 8.44%.	Maintain current equitable hiring practices to preserve the strong gender balance.
Performance Culture	69.47% are high performers. However, self-ratings are consistently 0.5 points higher than manager ratings.	Align manager rating calibration with employee self-assessments to reduce expectation gaps.
Tenure & Promotion	Average promotion lag is 3.44 years. Most attrition occurs around the 2.43-year mark.	Develop early-tenure retention programs targeting employees with under 3 years of service.

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